



Precondition Synthesis for DNNs with Statistical Guarantees

StatPre: A Select-and-Solve Framework

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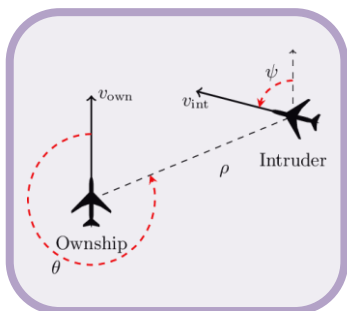
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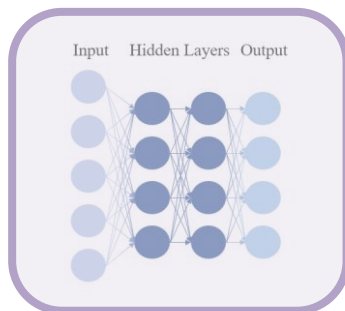
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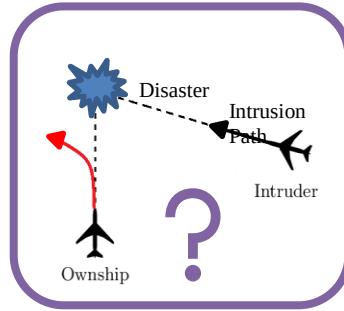
Background: Formal Specification of DNNs



Precondition



Model

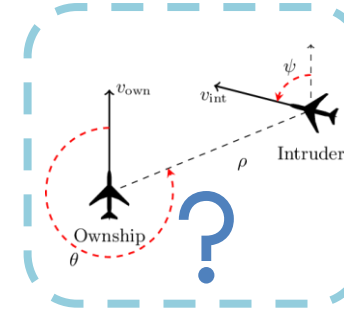


Postcondition

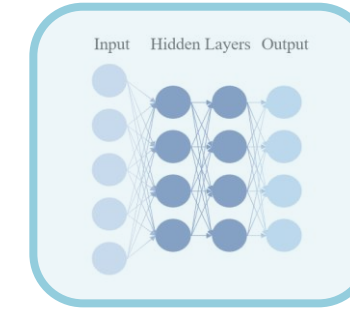
Postcondition Inference

Determines if all inputs within a domain (Precondition) satisfy property P (Post).

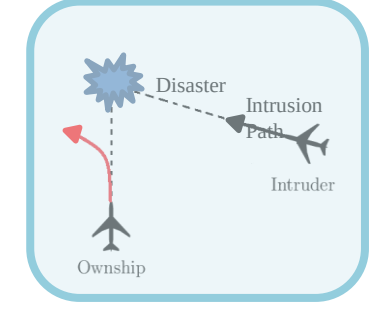
Limitation: Binary qualitative conclusion (Satisfied/Violated). Lacks quantitative analysis for subdomains.



Precondition



Model



Postcondition

Precondition Synthesis

Calculates the input space (Precondition) that ensures a property (Postcondition) holds.

Goal: To ...



Precondition Synthesis for DNNs Matters!

To improve the interpretability of DNNs themselves

- ✓ Solve which concrete partial input space meets the desired property
- ✓ Provide valuable specifications beyond the common benchmarks

To provide a reliability guarantee for DNN components at the system level

- ✓ Make quality assessments on trained DNNs for a controllable system integration
- ✓ Runtime monitor and make dynamic task allocation among modules, such as by using the Simplex architecture

Landscape of Precondition Synthesis

Exact Synthesis^[1,2]

Goal: Weakest Precondition.

Characterize all inputs ensuring property satisfaction.

Limitations:

- Complexity: At least **#P-hard**.
- Intractable for scaled DNNs.
- Limited to small ReLU nets with analytical representations.

Approximate Synthesis

Goal: Manage computational complexity to handle larger DNNs.

Scopes: Deterministic Over/Under-approximation.

- Current research limited to piecewise linear or low-dimensional scenarios.
- Most existing approximate methods rely on White-box encoding, leveraging intrinsic parameters and structures.

White-box Dependency^[3-6]

Heavily relies on trained parameters and piecewise linear structures (ReLU).

- Fails for general non-linear activations.
- Architecture-specific designs.

Gray-box/Verifier Bottleneck^[7]

Some methods treat DNNs as gray boxes but rely on formal verifiers for soundness.

- Limited by verifier capability.
- Struggle with high-dimensional inputs.

[1] K. Matoba, F. Fleuret, 2020. Exact Preimages of Neural Network Aircraft Collision Avoidance Systems. ML4Eng Workshop at NeurIPS.

[2] T. Nancy et al., 2022. An Analytical Approach to Compute the Exact Preimage of Feed-Forward Neural Networks. arXiv:2203.00438.

[3] D. Gopinath et al., 2019. Property Inference for Deep Neural Networks. ASE.

[4] S. Kotha et al., 2023. Provably Bounding Neural Network Preimages. NeurIPS.

[5] X. Zhang et al., 2024. Provable Preimage Under-Approximation for Neural Networks. TACAS.

[6] X. Zhang et al., 2025. PREMAP: A Unifying Preimage Approximation Framework for Neural Networks. JMLR.

[7] Z. Liu et al., 2024. Synthesizing Boxes Preconditions for Deep Neural Networks. ISSTA.

Motivation: Why StatPre?

We propose **StatPre**, a *select-and-solve* framework for automated precondition synthesis.

- ✓ **Black-box Paradigm:** No dependency on intrinsic structure or verifiers.
- ✓ **Theoretical Soundness:** Provides synthesized preconditions with formal Statistical Guarantees.
- ✓ **Versatility:** Enhances the ability to process complex, high-dimensional DNNs.

Optimization Goal

Maximize Weakness (Size)

&

Maximize Accuracy

Using a Box-based abstraction

Problem Formulation: Statistical Preconditions

Given a postcondition $\mathcal{P}: \varphi(f(x)) < 0$ for a trained DNN f over a bounded input domain $\mathcal{I} \in \mathbb{B}^n$, along with a significance level η and an error tolerance ϵ , how to synthesize **statistical guaranteed precondition predicates** that approximate the real preconditions as closely as possible and as weak as possible?

with confidence at least $1 - \eta$, $\mathbb{P}[\varphi(f(x)) < 0 \mid x \in \mathcal{B}] \geq 1 - \epsilon$

with confidence at least $1 - \eta'$, $\mathbb{P}[\varphi(f(x)) < 0 \mid x \in \mathcal{BM}] \geq 1 - \epsilon'$

a set $\mathcal{B} \subseteq \mathbb{R}^n$ is a n -dimensional **Box** $\mathcal{B}(x_0, \lambda) = \{x \in \mathbb{R}^n \mid x = x_0 + \lambda \odot v, \|v\|_\infty \leq 1, v \in \mathbb{R}^n\}$

a set $\mathcal{BM} \subseteq \mathbb{R}^n$ conforms to **Boxes** iff there exist n -dimensional boxes $\mathcal{B}_1, \dots, \mathcal{B}_k$ such that $\mathcal{BM} = \bigcup_{i=1}^k \mathcal{B}_i$

StatPre: Select-and-Solve Framework

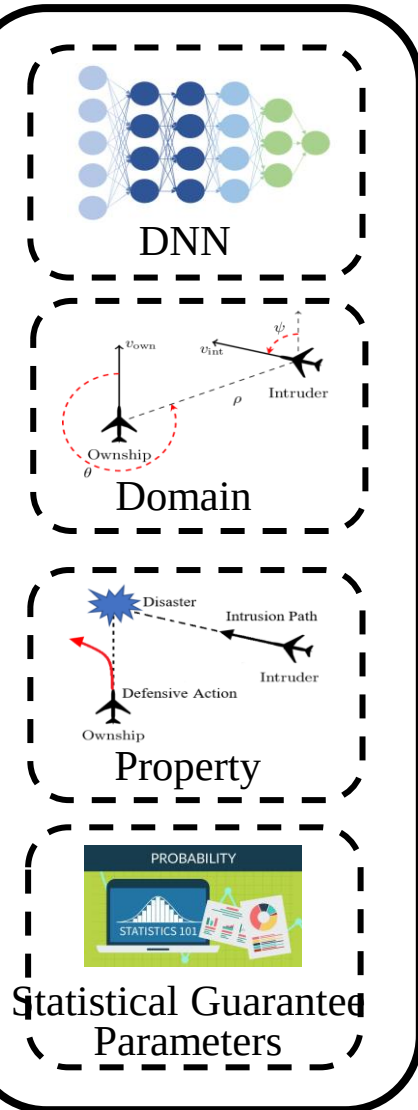
Center Selection

To select
a set of centers
with high potential to
better localize the
real preconditions

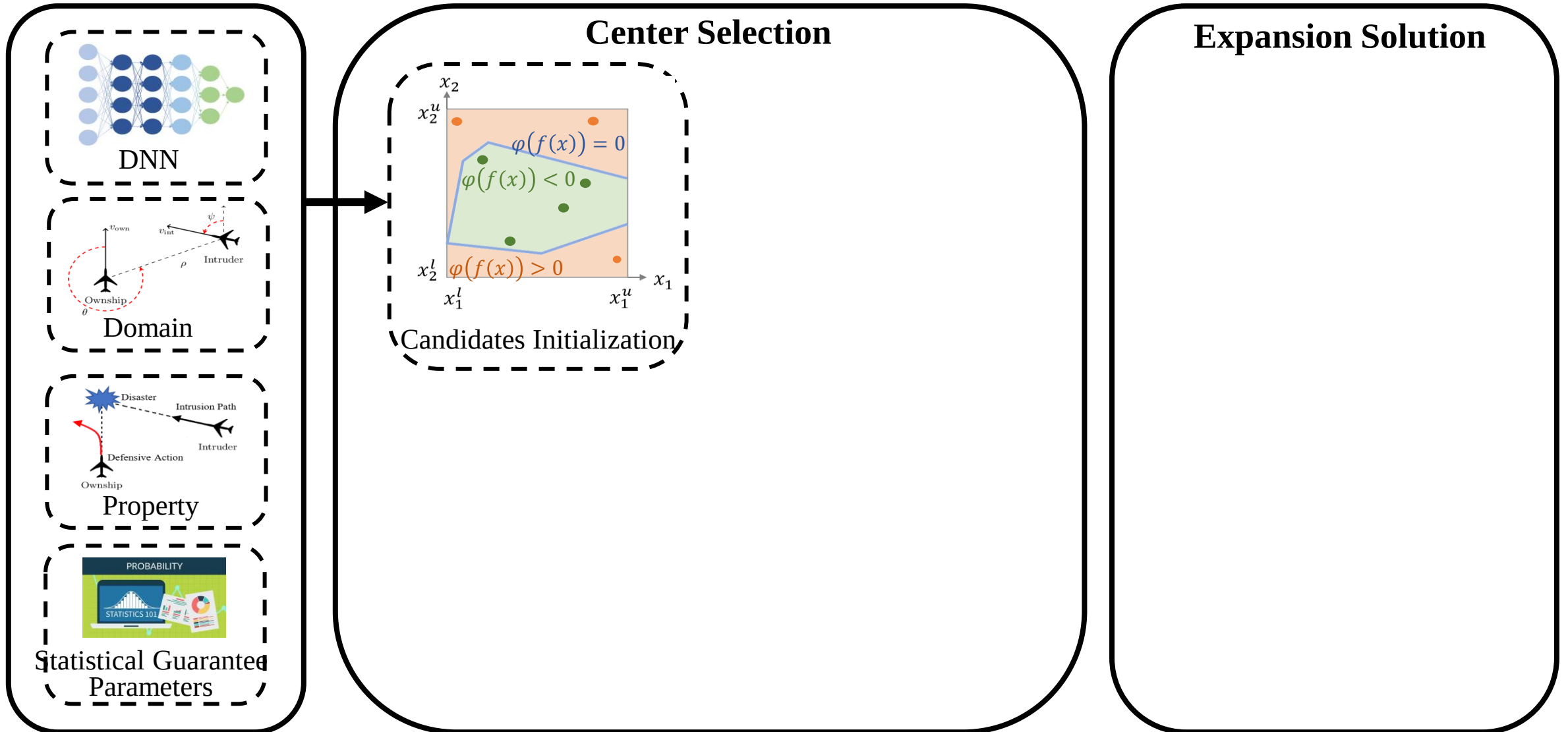


Expansion Solution

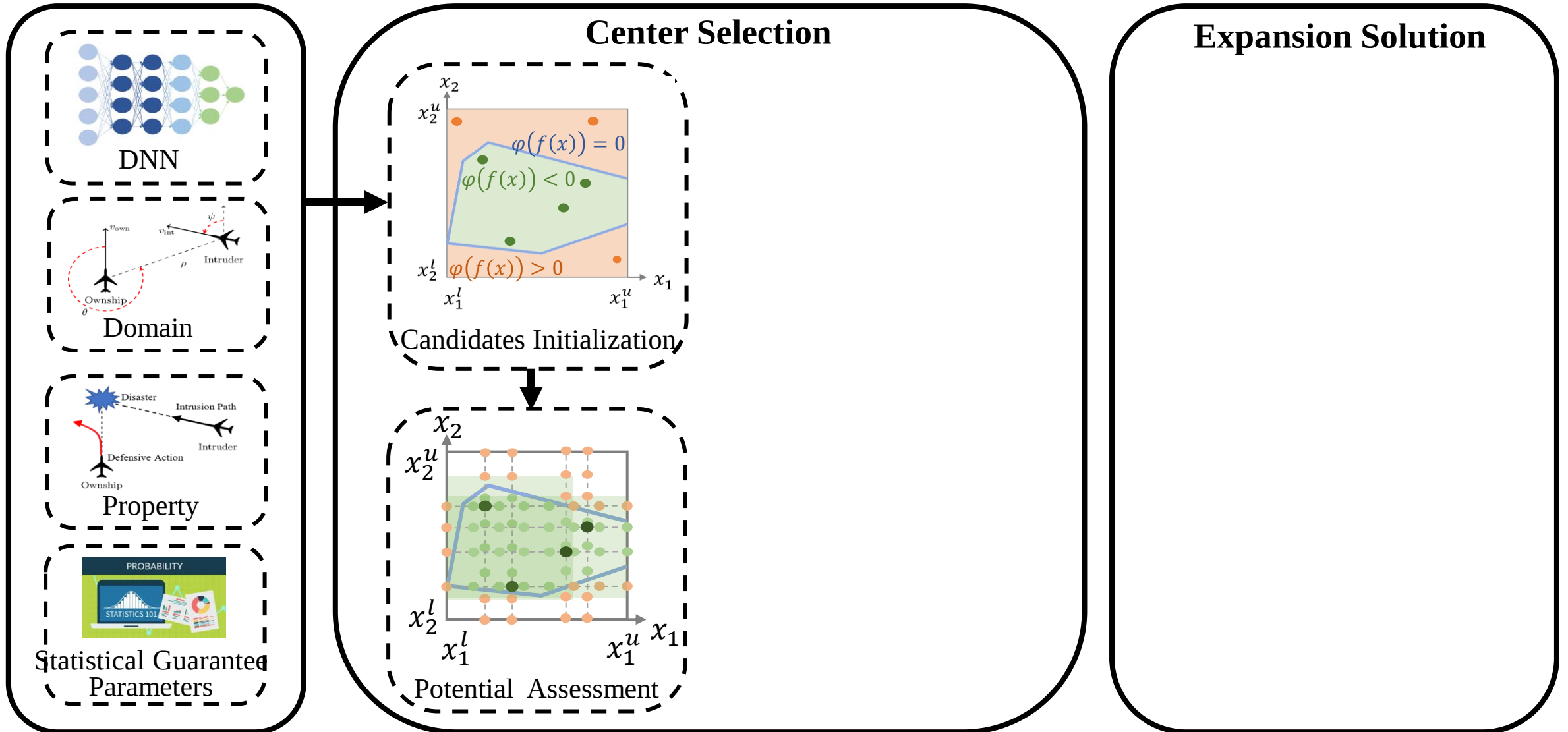
To expand
around each center
to form a precondition
as weak and
accurate as possible



StatPre: Select-and-Solve Framework



StatPre: Select-and-Solve Framework

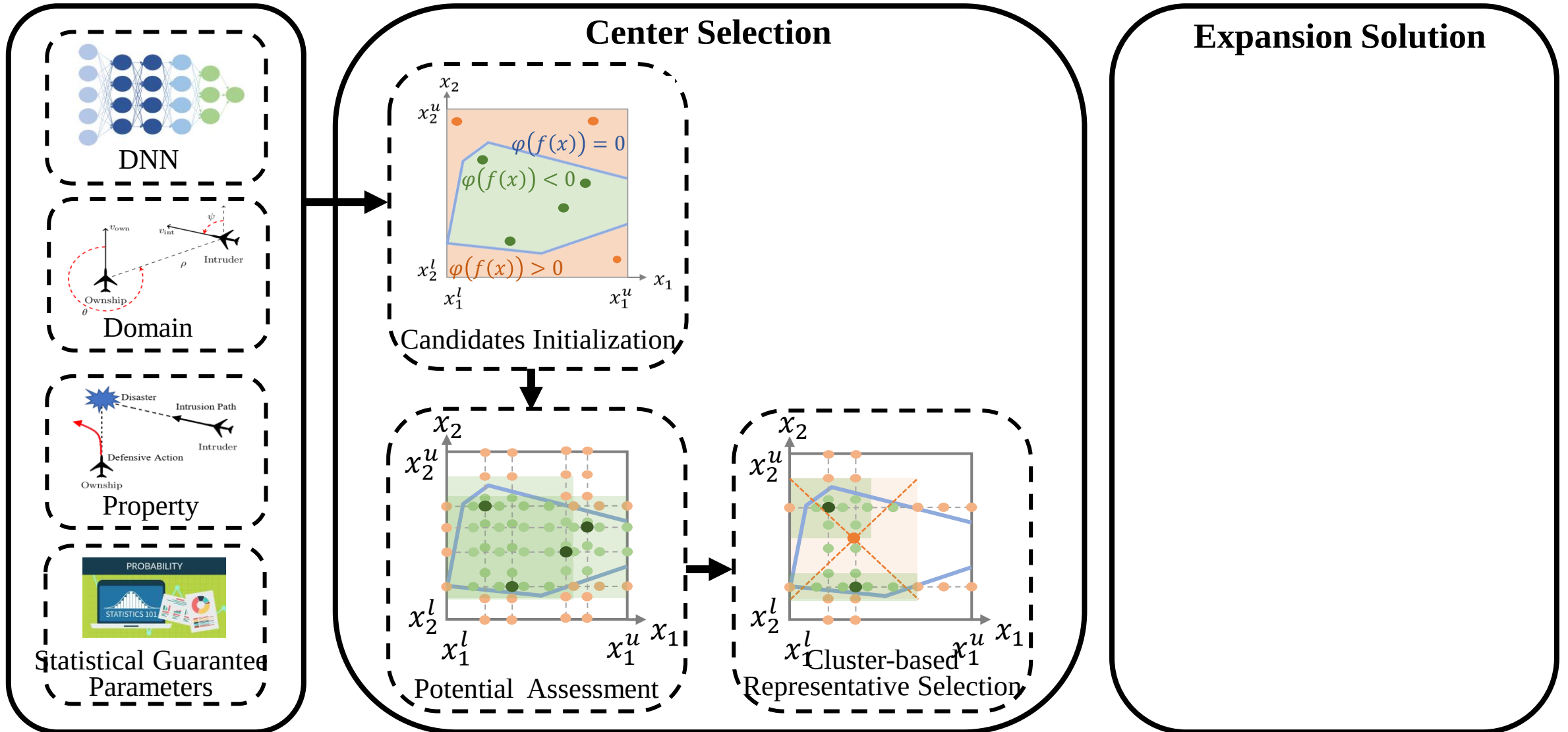




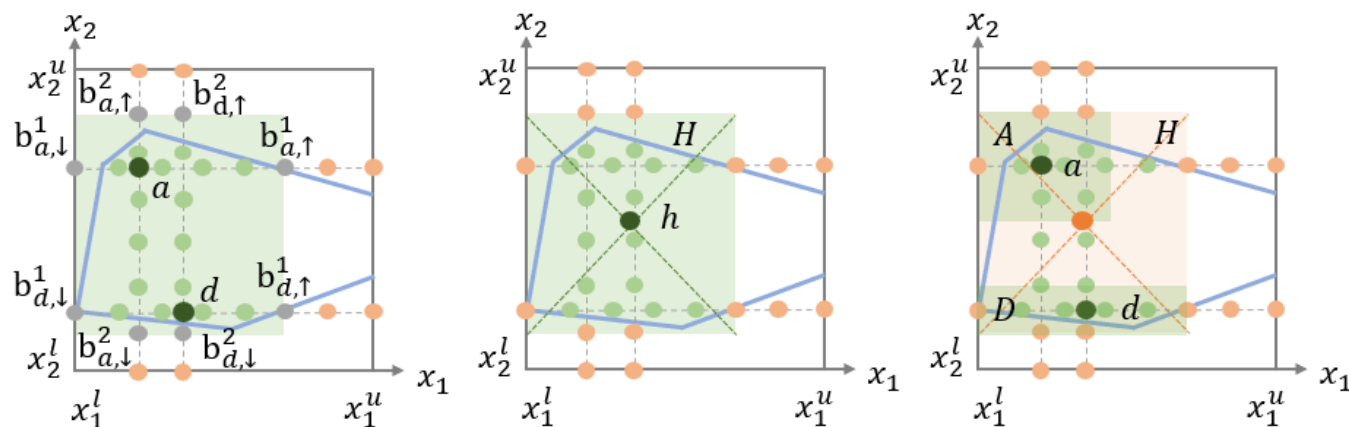
Efficiently approximates the connected potential region around candidates by searching for decision boundaries.

- ✓ Alleviates exponential costs of traditional grid sampling in high-dimensional domains.

StatPre: Select-and-Solve Framework



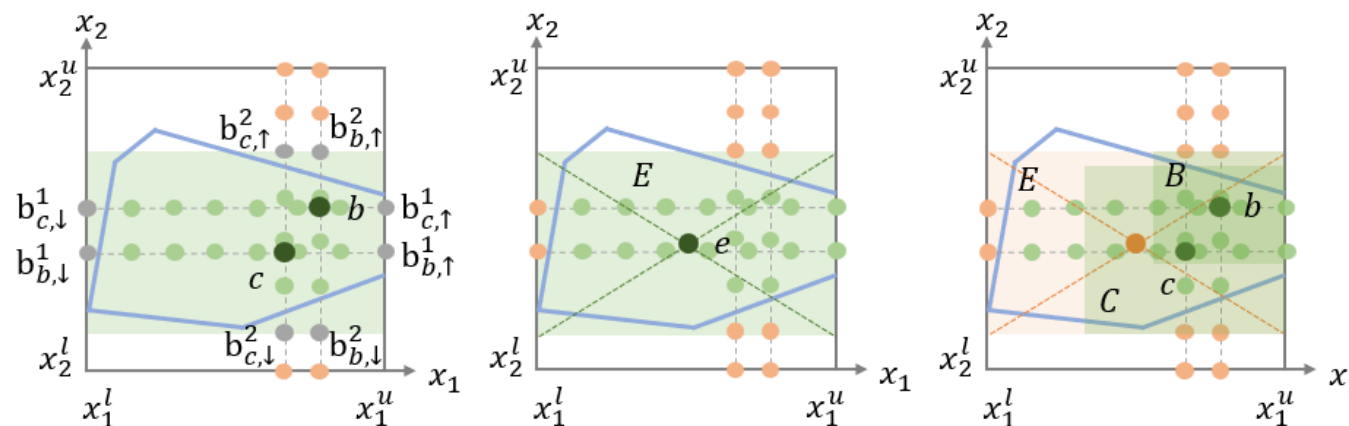
Cluster-Based Representative Selection



(a) Cluster a and d

(b) Symmetric Point h Valid

(c) Symmetric Point h Invalid



(d) Cluster b and c

(e) Symmetric Point e Valid

(f) Symmetric Point e Invalid

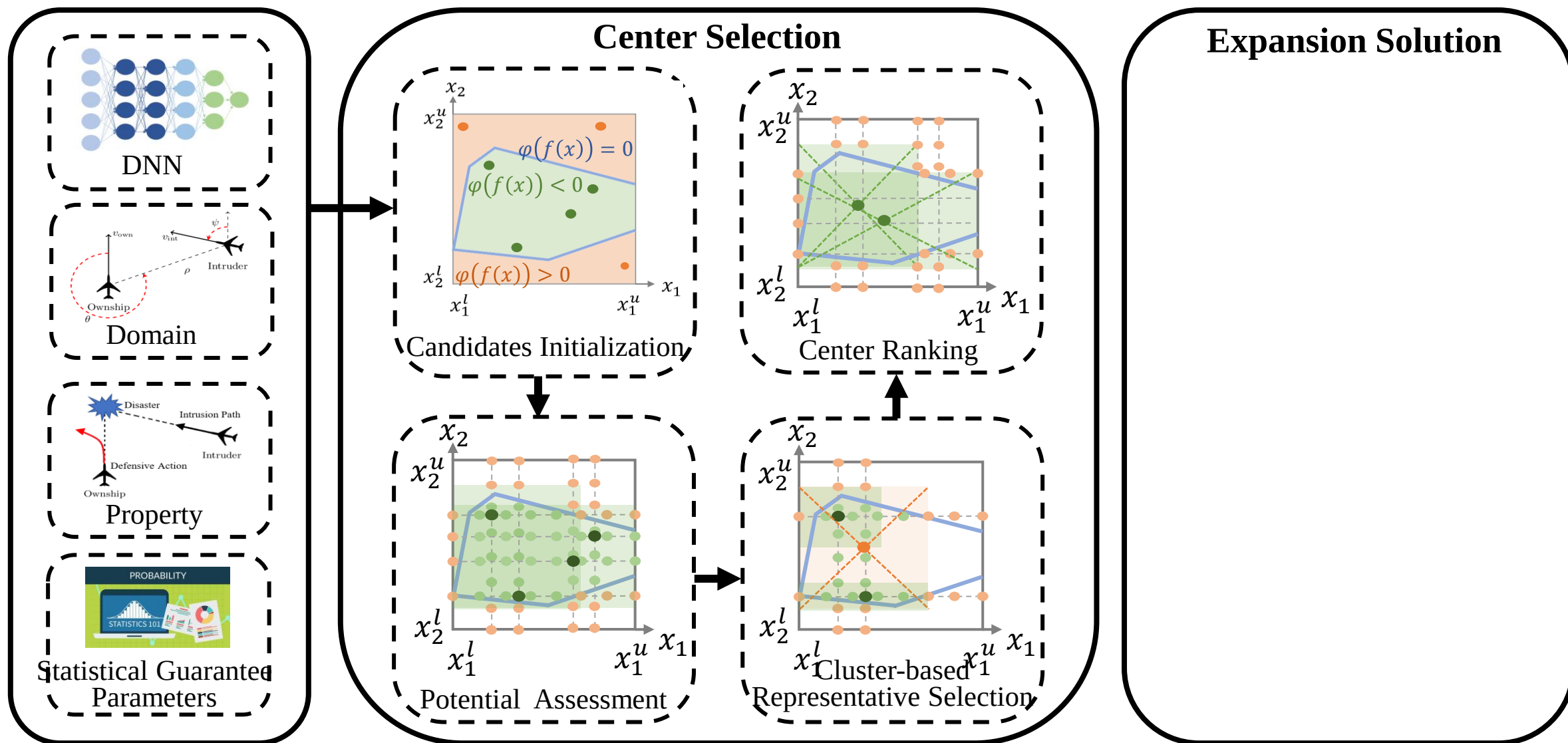
Identifying Core Regions

Clusters candidates sharing similar spatial structures to find points farther from decision boundaries.

Representative Selection: Prefer the center-symmetric point of the predicted connected subspace.

- ✓ Reduces bias from initial random samples.
- ✓ Offers greater expansion potential for Box synthesis.

StatPre: Select-and-Solve Framework



Expansion Solution

$$\begin{aligned} \max_{\lambda} \quad & \text{vol}(\lambda) \\ \text{s.t.} \quad & \varphi(f(x)) < 0 \text{ for all } x \in \mathcal{B}(x_0, \lambda). \end{aligned}$$



$$\begin{aligned} \max_{\lambda} \quad & \text{vol}(\lambda) \\ \text{s.t.} \quad & \varphi(f(x_0 + \lambda \odot v)) < 0 \text{ for all } v \text{ with } \|v\|_{\infty} \leq 1. \end{aligned}$$



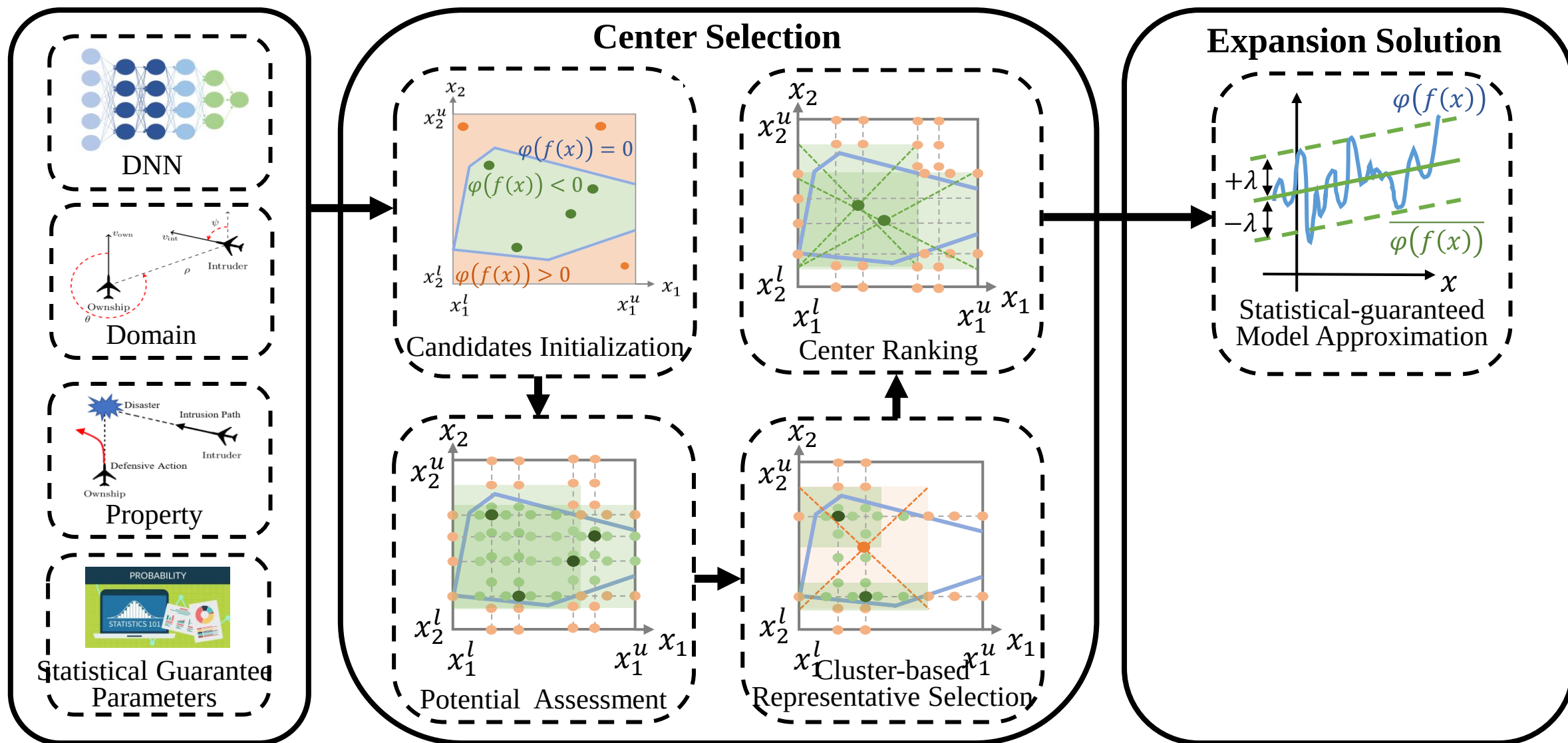
?

each tuple (x_0, λ, v) uniquely determines

$\varphi(f(\cdot))$ is a non-convex function with no tractable closed form

- × black-box nature of DNN
- × non-convexity introduced by activations

StatPre: Select-and-Solve Framework

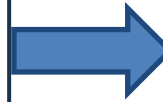


Statistical Model Approximation

$$\begin{aligned} & \max_{\lambda} \text{vol}(\lambda) \\ & \text{s.t. } \varphi(f(x)) < 0 \text{ for all } x \in \mathcal{B}(x_0, \lambda). \end{aligned}$$



$$\begin{aligned} & \max_{\lambda} \text{vol}(\lambda) \\ & \text{s.t. } \varphi(f(x_0 + \lambda \odot v)) < 0 \text{ for all } v \text{ with } \|v\|_{\infty} \leq 1. \end{aligned}$$



With confidence at least $1-\eta$,
the following holds within $B(x_0, \lambda)$

$$\mathbb{P} \left[|\varphi(f(x)) - (\varphi(f(x_0)) + a^{\top} \lambda + b^{\top} v + c)| \leq \mu \right] \geq 1 - \epsilon$$

PAC Model Approximation

Approximates $\phi(f(\cdot))$ as a linear model
within $B(x_0, \lambda)$ (potential domain)

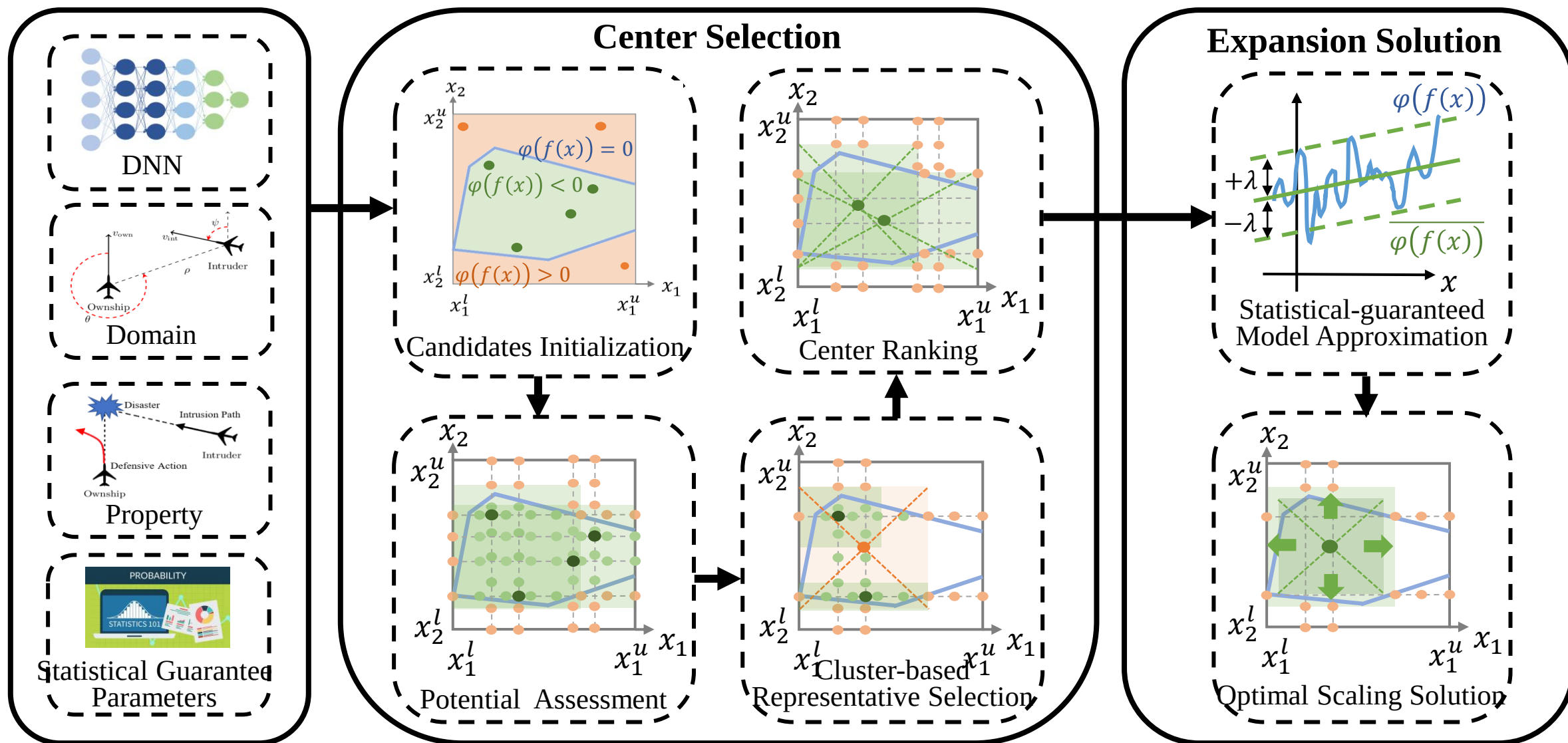
$$\varphi(f(x_0 + \lambda \odot v)) \approx \varphi(f(x_0)) + a^{\top} \lambda + b^{\top} v + c$$

convex optimization over i.i.d. samples,
with a sample count n adhering to the
Senoria-based sample complexity.

$$\begin{aligned} & \min_{a, b, c, \mu} \mu \\ & \text{s.t. } \varphi(f(x_0 + \bar{\lambda} \odot v^{(i)})) - \sigma(x_0, \bar{\lambda}, v^{(i)}) \leq \mu, \\ & \quad \varphi(f(x_0 + \bar{\lambda} \odot v^{(i)})) - \sigma(x_0, \bar{\lambda}, v^{(i)}) \geq -\mu, \\ & \quad l_a^d \leq a_d \leq u_a^d, \quad l_b^d \leq b_d \leq u_b^d, \quad d = 1, \dots, n, \\ & \quad l_c \leq c \leq u_c, \quad 0 \leq \mu \leq u_{\mu}, \end{aligned}$$



StatPre: Select-and-Solve Framework



Optimal Scaling Solution

$$\begin{aligned} \max_{\lambda} \quad & \text{vol}(\lambda) \\ \text{s.t.} \quad & \varphi(f(x)) < 0 \text{ for all } x \in \mathcal{B}(x_0, \lambda). \end{aligned}$$



$$\begin{aligned} \max_{\lambda} \quad & \text{vol}(\lambda) \\ \text{s.t.} \quad & \varphi(f(x_0 + \lambda \odot v)) < 0 \text{ for all } v \text{ with } \|v\|_{\infty} \leq 1. \end{aligned}$$

With confidence at least $1-\eta$,
the following holds within $B(x_0, \lambda)$

$$\mathbb{P} \left[|\varphi(f(x)) - (\varphi(f(x_0)) + a^{\top} \lambda + b^{\top} v + c)| \leq \mu \right] \geq 1 - \epsilon$$



Uses Hölder's Inequality to characterize worst-case impact of variables efficiently.

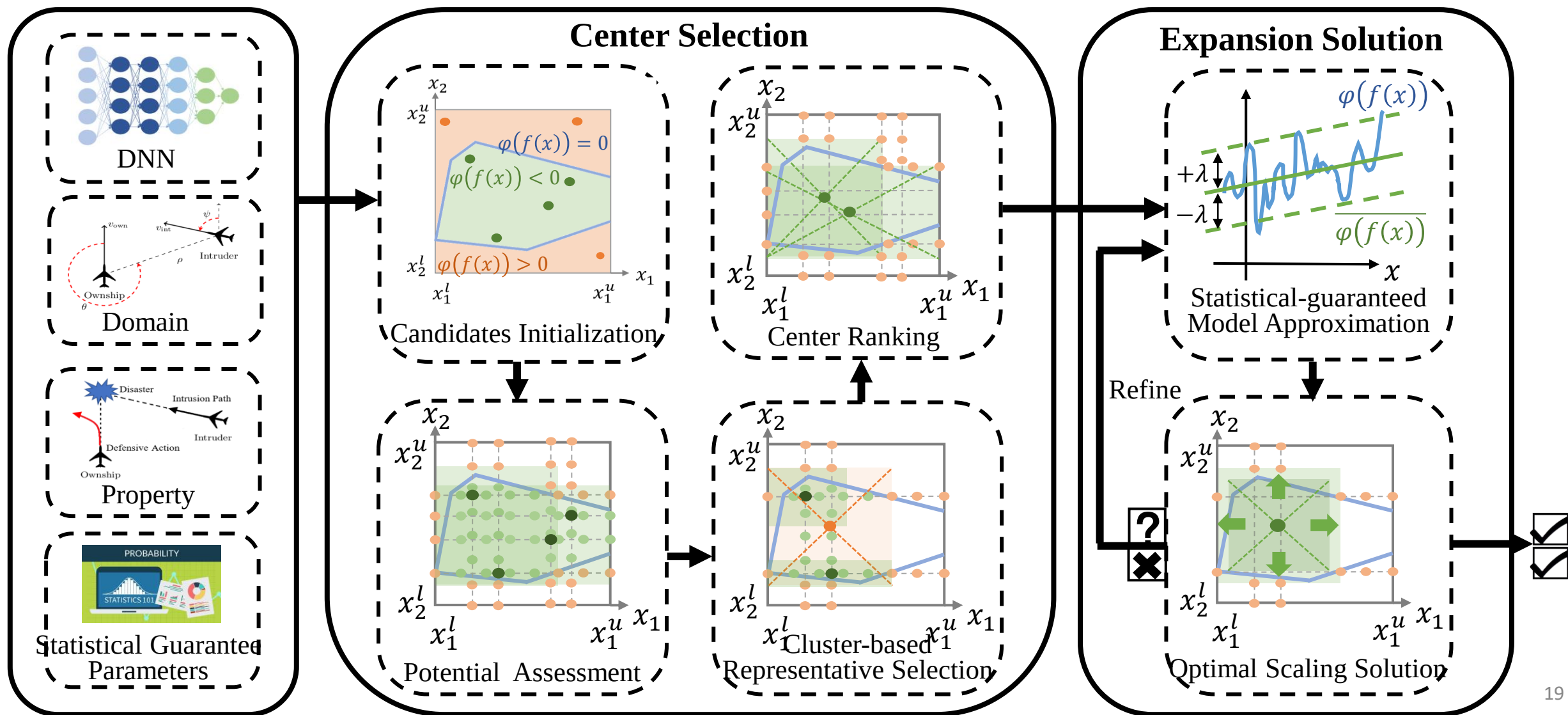
$$\begin{aligned} \min_{\lambda} \quad & -\sum_{i=1}^n \log \lambda_i \\ \text{s.t.} \quad & a^{\top} \lambda + \|b\|_1 + \varphi(f(x_0)) + c + \mu < 0 \\ & \text{for all } \mathbf{0} < \lambda \leq \bar{\lambda}. \end{aligned}$$



Converts the expansion problem into a convex optimization

$$\begin{aligned} \min_{\lambda} \quad & -\sum_{i=1}^n \log \lambda_i \\ \text{s.t.} \quad & (\varphi(f(x_0)) + a^{\top} \lambda + b^{\top} v + c) + \mu < 0 \\ & \text{for all } v \in \mathbb{B}_{\infty}^n(1) \text{ and } \mathbf{0} < \lambda \leq \bar{\lambda}. \end{aligned}$$

StatPre: Select-and-Solve Framework



Statistical Guarantees: Single to Multi-Boxes

Guarantees for a Single Box

Proposition 3.6. With confidence at least $1 - \eta$, we have $\mathbb{P}[\varphi(f(x)) < 0 \mid x \in \mathcal{B}(x_0, \lambda^*)] \geq 1 - \epsilon$.

Guarantee for Multi-Boxes (Union)

Proposition 3.7. With confidence at least $1 - m\eta$, we have $\mathbb{P}[\varphi(f(x)) < 0 \mid x \in \mathcal{BM}] \geq 1 - \epsilon \cdot \frac{\sum_{i=1}^m \text{vol}(\mathcal{B}_i)}{\text{vol}(\mathcal{BM})}$

Corollary 3.8. Under Proposition 3.7, the effective error parameter $\epsilon' = \epsilon \cdot \frac{\sum_{i=1}^m \text{vol}(\mathcal{B}_i)}{\text{vol}(\mathcal{BM})}$ satisfies $\epsilon \leq \epsilon' \leq m\epsilon$.

Key Relationship

Proposition 3.9. Higher confidence and correctness in the guarantees for individual box preconditions lead to higher confidence and correctness in the statistical guarantee for the union of boxes.

Benchmarks

Evaluate StatPre on 15 models with 27 properties across 6 benchmarks:

Benchmark	#Dim	#Prop	#Model	DNN Type	Application
Syn	2	10	1	FC.+ReLU	Classification
CartPole	4	2	1	FC.+ReLU	Reinforcement Learning
ACASXu	5	4	6	FC.+ReLU	Airborne Collision Avoidance
CifarBiasField	16	5	5	Conv.+ReLU	Image Classification
NN4Sys	154	2	1	Complex.+Sigmoid	Learned Cardinality Estimation
MNIST	784	4	1	FC.+ReLU	Image Classification

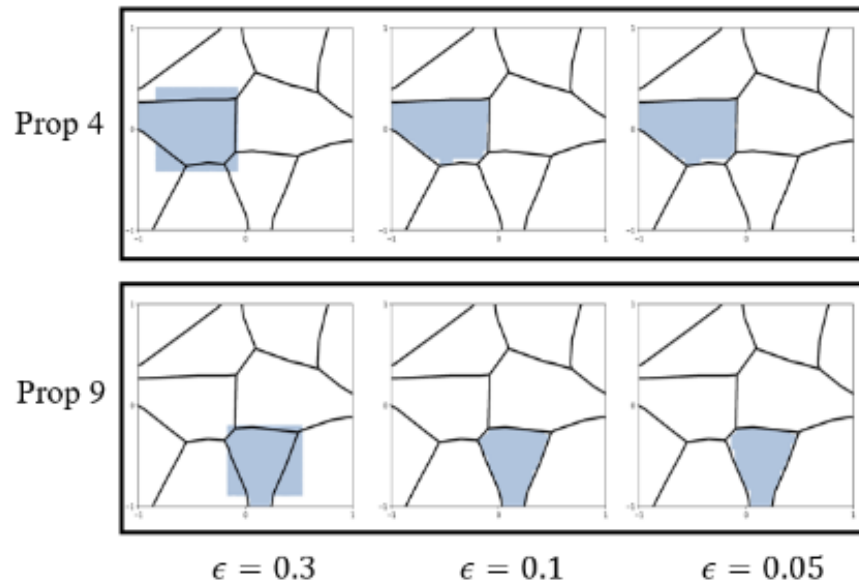
High-dimensional non-ReLU complex-structured DNNs.

Evaluation Metrics

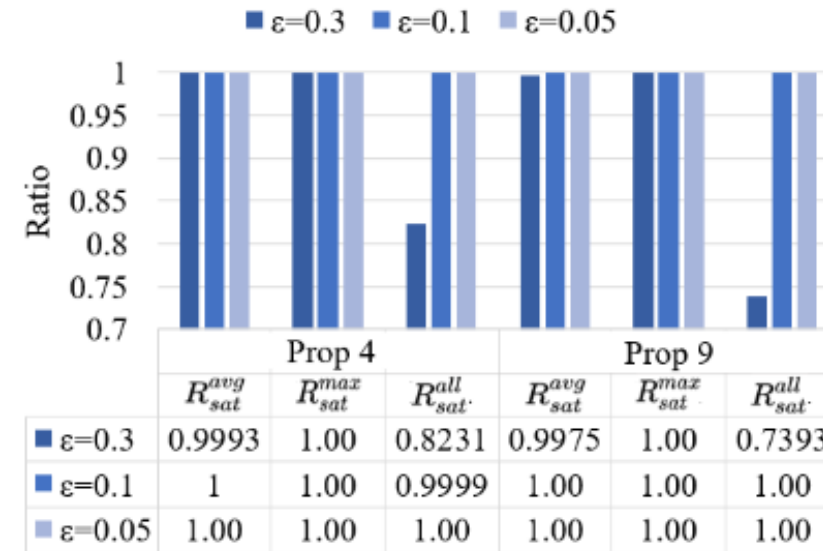
Category	Goal	Scope	Metric	Formula / Method
Accuracy	Measure how well the synthesis approximates property $\mathcal{P}$	Individual Box ($\mathcal{B}$)	$R_{\text{sat}}^{\text{avg}}, R_{\text{sat}}^{\text{max}}$	$R_{\text{sat}} \approx \frac{1}{k} \sum_{i=1}^k \mathbf{1}_{f(x_i) \in \mathcal{P}, x_i \in \mathcal{B}}$
		Union of Boxes ($\mathcal{BM}$)	$R_{\text{sat}}^{\text{all}}$	$R_{\text{sat}} \approx \frac{1}{k} \sum_{i=1}^k \mathbf{1}_{f(x_i) \in \mathcal{P}, x_i \in \mathcal{BM}}$
	Reliability Enhancement	Accuracy Check	$\alpha\beta$ -CROWN / PGD Attack	Formal verification and adversarial attacks to handle high-dimensional sampling gaps.
Weakness	Measure the coverage/size of the synthesized solution	Individual Box ($\mathcal{B}$)	$r_{\text{avg}}, r_{\text{max}}$	$r = \frac{1}{n} \sum_{i=1}^n \lambda_i \text{ (Geometric average diameter)}$
		Union of Boxes ($\mathcal{BM}$)	R_{cov}	$R_{\text{cov}} \approx \frac{1}{k} \sum_{i=1}^k \mathbf{1}_{x_i \in \mathcal{BM}, x_i \in \mathcal{J}}$

RQ1: Practical Reliability

? Can StatPre synthesize non-trivial preconditions with statistical guarantees that closely approximate the real preconditions in practice?



(a) Illustrative Examples on Syn



(b) Statistical View of Examples

Visualization and Data Representation on Syn

RQ1: Practical Reliability

? Can StatPre synthesize non-trivial preconditions with statistical guarantees that closely approximate the real preconditions in practice?

Model	Property	Single Box		Box Union		Verification	
		$R_{\text{sat}}^{\text{avg}}$	$R_{\text{sat}}^{\text{max}}$	#Box	$R_{\text{sat}}^{\text{all}}$	$\exists_{\text{pass}}^{\text{verify}}$	#pass
<i>syn</i>	<i>prop₄</i>	0.99999	1.0	238	0.99991	✓	236
<i>syn</i>	<i>prop₉</i>	1.0	1.0	169	1.0	✓	166
<i>cartpole</i>	<i>prop₁</i>	0.91089	0.99998	181	0.68886	✗	0
<i>cartpole</i>	<i>prop₂</i>	0.92024	0.99993	191	0.72942	✗	0
<i>acasxu_1_8</i>	<i>prop₂</i>	0.91715	1.0	379	0.89189	✓	22
<i>acasxu_3_6</i>	<i>prop₃</i>	0.99732	1.0	309	0.99614	✓	1
<i>acasxu_1_2</i>	<i>prop₄</i>	0.97137	1.0	13	0.96623	✗	0

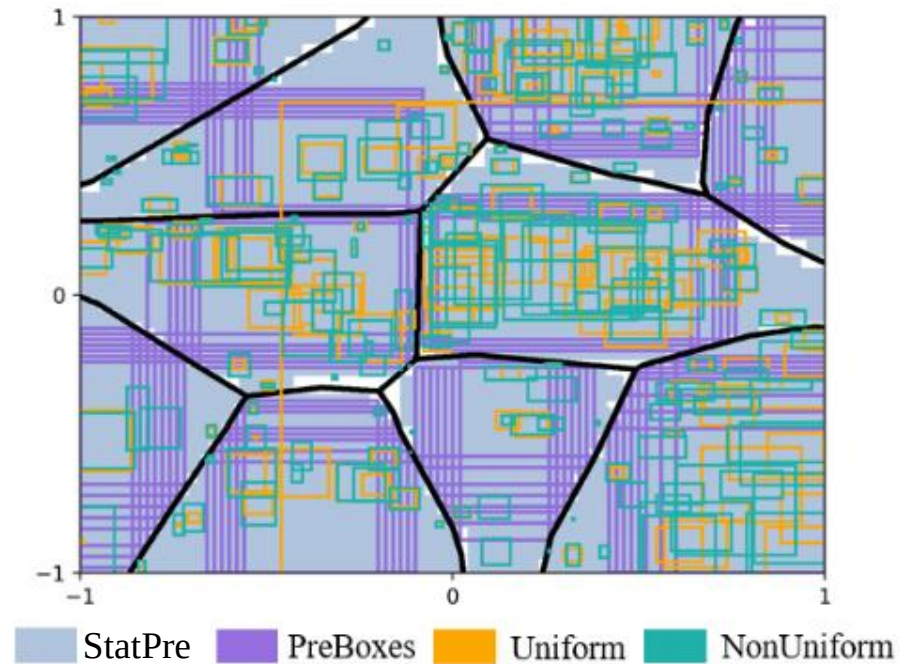
Guarantees Evaluation on Syn, Cartpole and ACASXu



The satisfaction ratio of the synthesized preconditions in practical tests typically exceeded the theoretical statistical thresholds, proving the framework's reliability.

RQ2: Comparison

? How does StatPre compare to existing approaches in terms of synthesis quality and efficiency? Specifically, **does it achieve broader precondition coverage than deterministic methods?**



Comparison with Uniform, NonUniform, and PreBoxes on Syn

RQ2: Comparison

? How does StatPre compare to existing approaches in terms of synthesis quality and efficiency? Specifically, **does it achieve broader precondition coverage than deterministic methods?**

Model	Property	Uniform			NonUniform			PreBoxes			StatPre				Time(s)
		r_{avg}	r_{max}	R_{cov}	r_{avg}	r_{max}	R_{cov}	r_{avg}	r_{max}	R_{cov}	r_{avg}	r_{max}	R_{cov}	R_{sat}^{avg}	
<i>cartpole</i>	<i>prop₁</i>	0.602	0.644	0.00103	0.160	0.162	0.0	56.776	69.276	0.751	68.484	93.327	0.818	0.886	2.59
<i>cartpole</i>	<i>prop₂</i>	0.526	0.644	0.0	0.157	0.162	0.0	56.776	60.942	0.642	63.063	84.262	0.694	0.918	1.76
<i>acasxu_1_8</i>	<i>prop₂</i>	4208.24	4208.26	0.0	4208.24	4208.28	0.0	8716.86	11053.63	0.250	10346.658	12633.34	1.0	0.922	43.62
<i>acasxu_1_6</i>	<i>prop₃</i>	4635.39	5289.97	0.0013	4896.96	5692.61	0.0018	10061.17	11113.79	0.316	10978.096	12628.65	0.971	0.884	46.60
<i>acasxu_3_6</i>	<i>prop₃</i>	NaN	4725.06	0.00004	NaN	5480.24	0.00004	10011.17	11083.63	0.113	11927.525	12633.31	1.0	0.997	47.68
<i>acasxu_1_2</i>	<i>prop₄</i>	NaN	4234.71	0.0	NaN	4367.75	0.0	0.0	0.0	0.0	6476.713	11907.69	0.025	0.991	56.57

Comparison with Uniform, NonUniform and PreBoxes on Cartpole and ACASXu



Under the high-confidence, low-error configurations to approximate the ground truth, StatPre synthesizes larger regions (broader coverage) than deterministic synthesis work (Unifor/NonUniform/PreBoxes).

RQ2: Comparison

? How does StatPre compare to existing approaches in terms of synthesis quality and efficiency? Specifically, **what additional benefits does it offer?**

Model	Property	R_{cov}	$R_{\text{sat}}^{\text{avg}}$	$\# \text{Box}$	$R_{\text{sat}}^{\text{all}}$	$\exists_{\text{pass}}^{\text{verify}}$	$\# \text{Pass}$	Time(s)
<i>nn4sys_mscn_2048d</i>	<i>prop</i> ₁	1.0	0.9692	45	0.9513	✓	39	282.308
<i>nn4sys_mscn_2048d</i>	<i>prop</i> ₂	1.0	0.9624	45	0.9419	✓	37	271.423
<i>cifarbiasfield_0</i>	<i>prop</i> ₀	0.9997	1.0	1	1.0	✓	1	59.335
<i>cifarbiasfield_1</i>	<i>prop</i> ₁	0.9998	1.0	1	1.0	✓	1	59.039
<i>cifarbiasfield_2</i>	<i>prop</i> ₂	0.9997	1.0	1	1.0	✓	1	60.354
<i>cifarbiasfield_3</i>	<i>prop</i> ₃	0.9997	1.0	1	1.0	✓	1	59.150
<i>cifarbiasfield_4</i>	<i>prop</i> ₄	0.9998	1.0	1	1.0	✓	1	58.459

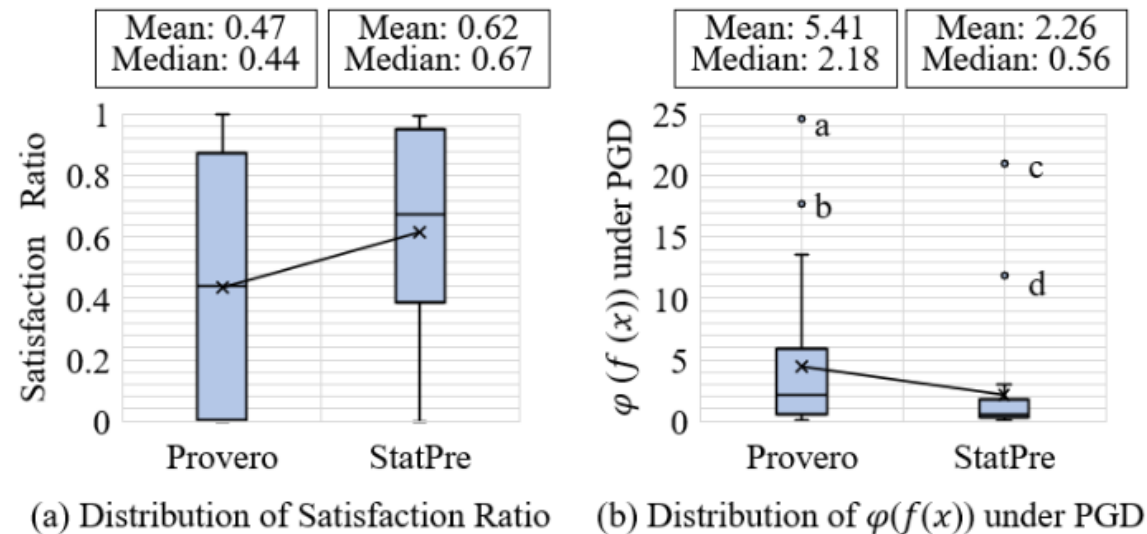
Precondition Synthesis on NN4Sys and CifarBiasField



StatPre can effectively handle input spaces up to 154 dimensions and manage complex non-ReLU structures, where many competitors fail due to model limitations or efficiency issues.

RQ2: Comparison

? How does StatPre compare to existing approaches in terms of synthesis quality and efficiency? Specifically, **does it provide more reliable preconditions than existing statistical techniques?**



Comparison with Provero on MNIST



Under the same theoretical guarantees by incorporating adversarial attacks, StatPre provides a more accurate approximation of the real precondition than statistical verification work (Provero).

RQ3: Ablation Study

? Do all functional factors in StatPre contribute as expected?

Model	Property	Cluster	r_{avg}	r_{max}
<i>acasxu_1_1</i>	<i>prop</i> ₁	✓	10993.837	12022.114
		✗	8115.823	11946.788
<i>acasxu_5_9</i>	<i>prop</i> ₃	✓	11803.229	12634.228
		✗	8224.655	12633.262
<i>cartpole</i>	<i>prop</i> ₂	✓	84.261	63.064
		✗	37.772	74.338
<i>cartpole</i>	<i>prop</i> ₁	✓	93.331	68.484
		✗	74.515	38.966

Ablation of Center Selection



Cluster-based representative selection improves coverage for individual statistical preconditions

Model	Property	Domain	μ_{avg}	μ_{max}	μ_{min}
<i>syn</i>	<i>prop</i> ₁	Potential	13.518	17.321	8.963
		Entire	37.254	38.697	34.862
<i>syn</i>	<i>prop</i> ₆	Potential	5.533	6.687	4.548
		Entire	30.639	34.234	26.916
<i>cartpole</i>	<i>prop</i> ₁	Potential	3.752	5.414	1.565
		Entire	6.218	7.646	4.807
<i>acasxu_1_2</i>	<i>prop</i> ₄	Potential	0.022	0.061	0.0002
		Entire	0.273	1.214	0.005

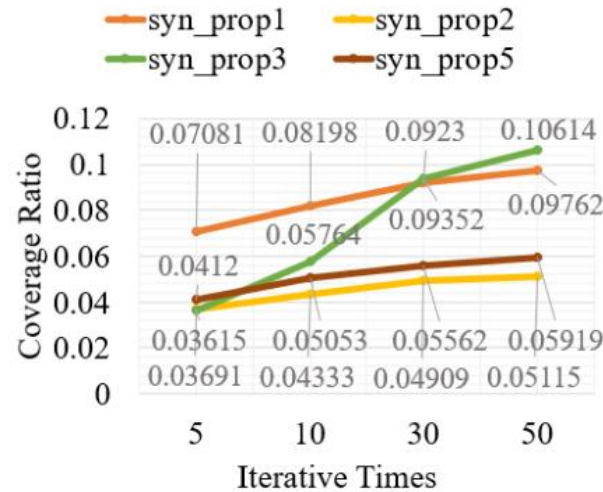
Ablation of Approximation Domain



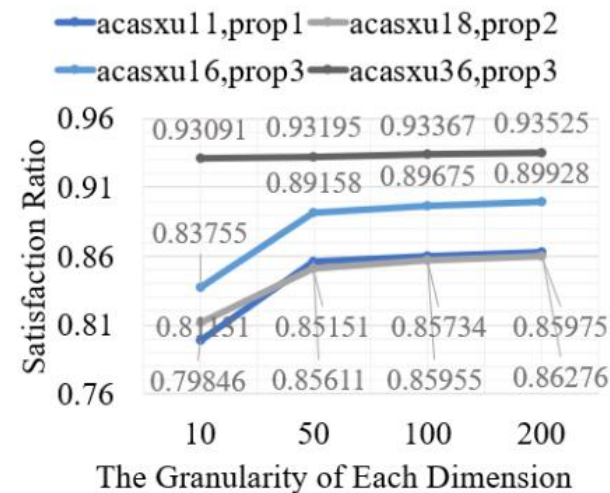
Modeling the potential neighborhood as a approximation domain enhance behavioral fidelity of DNN

RQ3: Ablation Study

? Do all functional factors in StatPre contribute as expected?



(a) Iterative Running



(b) Prediction Granularity

Impact of Typical Hyper-Parameter Settings



Iterative running contributes to generating a solution set with broader coverage.



Finer prediction granularity helps yield a more accurate representation of the potential space.

Summary & Takeaways

- ✓ Our **select-and-solve** framework enables DNN **black-box flexibility**.
- ✓ StatPre automatically synthesizes **statistically guaranteed preconditions** that are **as weak and as accurate to the real precondition as possible** via a **Box-based abstraction**

- ✓ StatPre effectively synthesizes preconditions with **broader coverage** than deterministic synthesis methods
- ✓ while **more accurately approximating** the ground truth compared to existing statistical verification technique.
- ✓ StatPre shows competitive capabilities to deal with **high-dimensional non-ReLU complex-structured DNNs**.



Precondition Synthesis for DNNs with Statistical Guarantees

StatPre: A Select-and-Solve Framework

Thanks!
Any Questions?

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